MAHMOUD GAMAL AHMED BAKHEET

Full Stack Developer (MERN / Next.js / NestJS)

Cairo, Egypt | +20 109 832 6602 | gamalgit@gmail.com | LinkedIn | GitHub | Portfolio

OBJECTIVE

Full Stack Developer skilled in MERN, Next.js, and NestJS, building production-style apps with clean architecture across JavaScript/TypeScript. Sole developer on a real-time broadcast graphics control system, with emphasis on secure, scalable, and performant systems delivered through continuous learning and best practices.

EDUCATION

B.Sc. in Computer Science — Arab Open University, Egypt — Graduated June 2026

EXPERIENCE

Full Stack Developer — TKDS MENA LLC — May 2026 – Aug 2026

Remote, from Egypt · part of TKDS Media LLC, USA

- Single-handedly designed and built a three-program CG system (Node.js/Express/Socket.io backend, React + Tailwind control panel, vanilla JS + GSAP overlay for vMix/OBS) supporting three live graphic types (score bug, lower thirds, leaderboards) within a three-month tenure — replacing a fully manual workflow and cutting graphic updates from a multi-step process to a single click, ready for live broadcast use.
- Eliminated race conditions across concurrent operator actions by designing a WebSocket event layer with a serialized write queue over a file-based JSON store, enabling reliable real-time state sync across every connected client without a database.
- Delivered the CG system end-to-end as the only engineer on it, alongside also building the official site for Mediacore2, the company's SaaS product — managing two concurrent deliverables independently.

System & Network Monitoring — Avnology — Apr 2024 – Aug 2024

Saudi Arabia-based company, remote from Egypt

- Detected and escalated production system and network incidents across shift-based monitoring, giving the technical team faster visibility into issues and supporting continuous uptime for client infrastructure.

PROJECTS

AI-Powered Brain Tumor Detection & Medical Consultation Platform — Graduation Project

React.js, Node.js, Express.js, Python (Flask, TensorFlow), MongoDB

- Built a full stack medical platform with 10+ features across doctor and patient dashboards, integrating an AI service that analyzes MRI scans and generates tumor-location heatmaps for doctor review.
- Applied role-based access control to separate doctor/patient permissions, secured medical data in transit and at rest, and decoupled the AI service from the main API so heavy inference never blocks request handling — a performance and scalability decision.

Football Scoreboard Overlay — Technical Assessment Project

NodeCG, Node.js, OBS Studio (Browser Source)

- Built a NodeCG bundle for a live football scorebug: a dashboard to control scores, match clock, team names, colors, and logos (via NodeCG's Asset System), synced instantly to a 1920×1080 OBS overlay.

Full-Stack Food Ordering App — Technical Assessment Project

Next.js, Express, MongoDB, TypeScript

- Delivered a bilingual (English/Arabic) food delivery app with an admin dashboard in a 24-hour assessment, using JWT auth with hashed credentials, i18n-driven RTL/LTR layout, and server-side rendering for faster load performance.

NestJS API Project — Personal Project

NestJS, TypeORM, PostgreSQL — modular backend applying dependency injection, DTO validation pipes, guards and interceptors, in a clean layered architecture separating controllers, services, and data access for long-term maintainability.

CERTIFICATIONS AND COURSES

Udemy: JavaScript | React.js, Next.js, TypeScript, Node.js, Express.js, Clean Code | NestJS

Self-study: HTML, CSS, Bootstrap, Tailwind CSS

TECHNICAL SKILLS

Languages & Frontend: JavaScript, TypeScript, SQL, React.js, Next.js, HTML5/CSS3, Tailwind CSS

Backend & Databases: Node.js, Express.js, NestJS, REST/GraphQL APIs, WebSockets, PostgreSQL, MongoDB, Prisma, TypeORM

Security & Architecture: JWT, OAuth, RBAC, Clean Architecture, Design Patterns, SOLID

Tools & Currently Learning: Jest, Git/GitHub, CI/CD — learning Redis, Docker, Kubernetes, System Design

LANGUAGES

Arabic — Native | English — Intermediate